

data gives the designer a better understanding of the device characteristics, to better judge the acceptability of the device to the application. Again, to run such tests, you need both a stimulus system like signal generator and an acquisition system like oscilloscope.

Verification. Verification is done to check whether a product meets a set of initial design requirements, specifications and regulations. In the development phase, verification procedures involve special tests on various modules. The results are modeled and the modeled results are reviewed or analysed to see whether the product complies with specifications. In the post-development phase, verification procedures involve regularly repeating tests devised specifically to ensure that the product continues to meet the initial design requirements, specifications and regulations as time progresses. A lot of development phase and post-development phase verification tests require a stimulus system like signal generator.

Type of signal generators

Signal generators are broadly divided into mixed-signal generators and logic sources, covering the whole range of signal-generation needs. Each of these types has its unique strengths that make it more or less suitable for specific applications.

Mixed-signal generators. These are designed to output waveforms with analogue characteristics. They produce various types of waveforms like sine, square/rectangular, sawtooth/triangular, step/pulse and various complex waves. These waveforms show the rounding and imperfections that are part of every real-world signal. Various parameters of these waveforms, such as amplitude, frequency, phase, DC offset and rise time/fall time, can be controlled in a mixed-signal generator. You can even create aberrations such as overshoot and add edge jitter, modulation and more.

Logic/digital sources. True digital sources, on the other hand, are designed to drive digital systems.

Unlike mixed-signal generators, these cannot generate analogue waveforms like sine or triangular waves. Such signal generators output binary pulse streams. Their features are optimised for computer bus needs and similar applications.

Analogue or digital signal generator

Most signal generators today are based on digital technology. Many can fulfil analogue requirements as well. Arbitrary waveform generators (AWGs) and function generators are primarily designed for analogue and mixed-signal applications. These instruments use sampling techniques to build and modify waveforms of almost any imaginable shape.

What to look for while buying one

This is actually a difficult question, and the selection is completely driven by your specific needs. A detailed understanding of your requirements and specifications mentioned in the datasheet can help you select the right signal generator for you. A signal generator produces various waveforms and its specifications are directly related to waveform properties.

Usually, signal generators produce electrical waves that repeat in a controllable manner. Each full repetition of a wave is known as 'period.' A waveform is actually a graphic representation of the wave's variation over time.

Amplitude. Amplitude is a measure of the voltage strength of the

waveform. Amplitude of an AC waveform periodically changes with time. Signal generators allow you to set a voltage range over which the signal continuously fluctuates with a frequency selected by you. The various amplitudes of interest in a waveform are peak amplitude, peak-to-peak amplitude and RMS amplitude (peak amplitude/ $\sqrt{2}$), as indicated in Fig. 2.

The amplitude specification for a common signal generator is 10-20V (open-circuit) peak-to-peak with 50 ohms impedance. 10V peak-to-peak sine wave has an RMS voltage of $10/\sqrt{2}$, which implies that the generator will be able to deliver only $10/50 \sqrt{2}$ amperes of RMS current at peak output. If you need more current or voltage than what is specified above, you will have to use a suitable amplifier or go for specialised function generators. So carefully check whether this specification suits your requirement.

For frequency response testing, amplitude flatness is also important. Amplitude flatness means that the fixed amplitude does not vary with the change in frequency.

Frequency. This is the rate at which full waveform cycles occur. Frequency is measured in Hertz (Hz), also known as cycles per second. It is inversely related to the wavelength (shown in Fig. 3 as point 4) of the waveform, which is a measure of the distance between two similar peaks on adjacent waves. The higher the frequency of the waveform, the shorter the time period.

The frequency specification of a signal generator is the highest sine wave frequency that the generator is capable of delivering. Many signal generators will have a lower maximum frequency specification for other waveforms. Here the minimum frequency that the generator can deliver is also important. For some applications, you will need very low frequencies. Signal generators with 1-micro-hertz resolution are readily available.

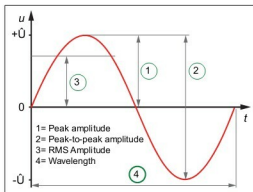


Fig. 2: Amplitude of an AC waveform